Physics 2300: Quantum Computing
Week 1

Course Outline
- posted on OWL Brightspace
  (switch to Outline)
- failure to read and follow course outline are NOT grounds for appeal

Why the hype about QC?
Why is it different from classical computing?

Classical Computing:
- manipulate bits represented as "0" and "1"
- if we measure a bit we get 0 or 1 <u>exclusively</u>
- a bit cannot be both 0 and 1 at the same time
- measuring a bit does not change its value
- all calculations are deterministic

Quantum Computing:
- manipulate <u>qubits</u> represented as "$|0\rangle$" and "$|1\rangle$"
- if we measure a qubit we get 0 or 1 <u>exclusively</u>
- a qubit CAN be BOTH $|0\rangle$ and $|1\rangle$ at the same time
- measuring a qubit forces it into $|0\rangle$ or $|1\rangle$ (kills any mix)
- calculations may be probabilistic

this can be good and bad

Why is this helpful?

eg. Most decryption algorithms are in some way equivalent to guessing a "key" which is an integer between 0 and 2^n, with $n = 128$ for strong encryption.

Classically, guessing the key requires sequentially trying all 2^n possibilities.
How long would this take for a <u>classical computer</u> that could try 1 possibility per clock cycle?

take $n = 128$, cycle time $= \frac{1}{5\,\text{GHz}} = 0.2$ ns

To try all 2^n possibilities requires

$$2^{128} \times 0.2 \text{ ns} = 3.4 \times 10^{38} \times 0.2 \times 10^{-9} \text{ s}$$
$$= 6.8 \times 10^{28} \text{ s}$$
$$= 2.16 \times 10^{21} \text{ years}$$
$$\approx 1.5 \times 10^{11} \times (\text{age of universe})$$

How long for a quantum computer?
As each qubit can be both $|0\rangle$ AND $|1\rangle$ simultaneously, with n qubits I can create ALL 2^n possibilities coexisting at the same time
$\Rightarrow$ This could take only 1 clock cycle ($\sim$ns) regardless of n (as long as I had enough qubits).

This is why the US military has a program monitoring cryptocurrencies. It is assumed that a "first use" of a fully functional QC would be to suck the value out of the cryptocurrency market.

The collapse of a qubit into $|0\rangle$ or $|1\rangle$ exclusively also allows an unbreakable <u>quantum</u> encryption.

All of Quantum Computing on One Slide

- **The good**
  - 2^n-way parallelism from n qubits
  - Possibility of exponential speedup for some problems
  - Some classically intractable problems can be made tractable
  - Some tractable problems can be solved asymptotically faster
  - Some problems can be solved exactly in the time it would take classically to solve them only probabilistically
- **The bad**
  - Quantum computation is extremely I/O bottlenecked: only n bits of input and n bits of output relative to 2^n-way parallelism
    - Can you think of a problem that reads a single 32-bit number, performs sequences of 4,294,967,296 concurrent operations on that number, and writes a single 32-bit number?
  - Limited applicability—note the use of "*some* problems" above
  - Programming is extremely difficult: requires expertise in linear algebra, computer science, and quantum physics as well as knowledge of prior algorithms and innate creativity
- **The ugly**
  - Contemporary quantum computers provide too few qubits even to represent most interesting problems
  - Current qubit quality is extremely low: unlikely to produce correct answers for more than handful of qubits running for more than a handful of time steps

Paikin & Rieffel, 2022

We will try to develop this expertise and knowledge in this course

Improvements in hardware are coming at a steadily increasing rate. Already things look much more promising than they did in 2022.

QC will almost certainly become a practical tool within the next 5-10 years. You need to be ready.

Linear Algebra over $\mathbb{C}^n$

notation: $\mathbb{Z} \equiv$ set of integers
$\mathbb{R} \equiv$ set of real numbers
$\mathbb{C} \equiv$ set of complex numbers
$z \in \mathbb{C} \equiv z$ belongs to set of complex numbers
i.e. z is a complex number

$z \in \mathbb{C}$ can be written in cartesian form as
$z = x + iy$, where $\{x, y\} \in \mathbb{R}$ and $i^2 = -1$
$\mathrm{Re}(z) = x$, the real part of z
$\mathrm{Im}(z) = y$, the imaginary part of z

Note: Real numbers are a subset of $\mathbb{C}$

$z^* =$ complex conjugate of z
$= x - iy$
note that if $z, w \in \mathbb{C}$ that
$(z+w)^* = z^* + w^*$
$(zw)^* = z^* w^*$
and $z^* = z$ <u>if and only if</u> $z \in \mathbb{R}$

The norm, magnitude, modulus, or absolute value of z is $|z| = \sqrt{x^2 + y^2}$
(note: $|z| \in \mathbb{R}$, $|z| \geq 0$, $|zw| = |z||w|$)

- addition and multiplication: let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z_1 \cdot z_2 = (x_1 + iy_1)(x_2 + iy_2) = x_1x_2 + ix_1y_2 + ix_2y_1 + \underbrace{i^2}_{=-1} y_1y_2$$

$$= \underbrace{(x_1x_2 - y_1y_2)}_{\mathrm{Re}\,(z_1z_1)} + i \underbrace{(x_1y_2 + x_2y_1)}_{\mathrm{Im}\,(z_1z_2)}$$

$z^{-1} = \frac{1}{z} =$ Inverse of $z = \frac{z^*}{|z|^2}$

so that

$$\frac{z_1}{z_2} = z_1 z_2^{-1} = (x_1 + i y_1)\left(\frac{x_2}{x_2^2+y_2^2} - \frac{y_2}{x_2^2+y_2^2}\, i \right)$$

Euler's Formula: $e^{i\phi} = \cos\phi + i \sin\phi$

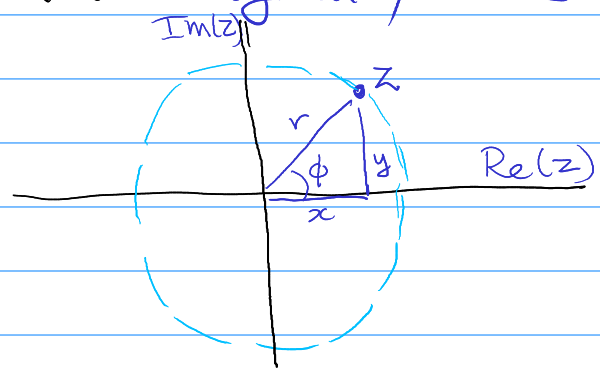
Polar or Exponential form

$$z = x + iy = re^{i\phi} = r(\cos\phi + i\sin\phi)$$
$$= \underbrace{r\cos\phi}_{x} + i\underbrace{r\sin\phi}_{y}$$

$\Rightarrow r = \sqrt{x^2+y^2} = |z|$, $\phi \equiv \mathrm{Arg}(z) =$ phase of z

$\cos\phi = \frac{x}{r}$, $\sin\phi = \frac{y}{r}$

$\Rightarrow$ any z can be thought of as a point in the complex plane with real axis x and imaginary axis y:



Note that the argument of a complex number ϕ is defined only up to integer multiple of 2π:

$$re^{i\phi} \equiv re^{i(\phi+2\pi n)}, \qquad n \in \mathbb{Z}$$

Examples:

$$i = e^{i\frac{\pi}{2}}, \qquad -1 = e^{i\pi}, \qquad 1 + \sqrt{3}i = 2e^{i\frac{\pi}{3}}$$

$$z^* = re^{-i\phi}, \qquad z_1 z_2 = r_1 r_2 e^{i(\phi_1+\phi_2)}, \qquad z^n = r^n e^{in\phi}$$

Beware: $\phi = \text{Arctan}\left(\frac{y}{x}\right)$ is <u>not</u> always correct as $\text{Arctan} \in \{-\frac{\pi}{2}, \pi/2\}$ so is always in <u>right</u> half plane. To decide whether or not to subtract π from $\text{Actan}(y/x)$ to get the phase, you should examine $\text{sign}(x)$ to decide if it is in the left half-plane.

Which form should you use?
Some computations, particularly multiplication & division are easier in polar form

eg. $z^n = r^n e^{in\theta}$

$$z_1 z_2 = r_1 r_2 e^{i(\phi_1 + \phi_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\phi_1 - \phi_2)}$$

$$z^* = r e^{-i\phi}$$

However, addition & subtraction are usually easier in cartesion form. (i.e. $z = x + iy$).

Complex Vector Spaces

A ket $|\psi\rangle$ is a "state vector" that describes a quantum system. $|\psi\rangle \in \mathbb{C}^n$, i.e. is a vector in n-dimensional linear complex vector space. We can represent $|\psi\rangle$ as a column vector,

$$|\psi\rangle \equiv \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}, \qquad z_i \in \mathbb{C},\ i = 1, 2, \cdots n$$

are complex numbers

Note: These behave like normal column vectors in linear algebra

$$\text{eg. } |\psi\rangle + |\varphi\rangle = \underbrace{\begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}}_{|\psi\rangle} + \underbrace{\begin{bmatrix} z_1' \\ z_2' \\ \vdots \\ z_n' \end{bmatrix}}_{|\varphi\rangle} = \begin{bmatrix} z_1 + z_1' \\ z_2 + z_2' \\ \vdots \\ z_n + z_n' \end{bmatrix}$$

$$a|\psi\rangle = a \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} = \begin{bmatrix} a z_1 \\ a z_2 \\ \vdots \\ a z_n \end{bmatrix}$$

A bra $\langle\psi|$ represents $|\psi\rangle$ in a dual (adjoint) space: as a row vector

$$\langle\psi| = \begin{bmatrix} z_1^* & z_2^* & \cdots & z_n^* \end{bmatrix}, \qquad \langle\psi| = \left(|\psi\rangle^{\overset{\text{transpose}}{T}}\right)^* = (|\psi\rangle)^{\dagger}$$

adjoint = complex conjugate transpose

Note: The elements of $\langle\psi|$ are the $\boxed{\text{complex conjugate}}$ of elements of $|\psi\rangle$

eg. $|\psi\rangle = \begin{bmatrix} 1+i \\ 0 \\ \sqrt{2} \end{bmatrix} \Rightarrow \langle\psi| = \begin{bmatrix} 1-i & 0 & \sqrt{2} \end{bmatrix}$

don't forget * changes $i \to -i$

$\langle\varphi| = \begin{bmatrix} \pi & -2 & -2i \end{bmatrix} \Rightarrow |\varphi\rangle = \begin{bmatrix} \pi \\ -2 \\ 2i \end{bmatrix}$

Basis Sets in $\mathbb{C}^n$

A basis set in $\mathbb{C}^n$ is a collection of n kets

$|v_1\rangle, \cdots, |v_n\rangle$

that

i) are linearly independent

i.e. $\sum_i a_i |v_i\rangle = 0 \Rightarrow a_i = 0$ for all i,

ii) and they produce a spanning set for $\mathbb{C}^n$

i.e. for any $|\psi\rangle \in \mathbb{C}^n$, there exists (complex number) coefficients $a_i \in \mathbb{C}$ such that

$$|\psi\rangle = \sum_i a_i |v_i\rangle = a_1|v_1\rangle + \cdots + a_n|v_n\rangle.$$

Note:

- the number of basis vectors is the dimension of the vector space
- given the basis, the set of coefficients a_i in the decomposition of any vector is unique
- the basis set is <u>not</u> unique

Example: $\mathbb{C}^2$ — a vector space corresponding to space (*i.e.*, the set all possible states) of a single qubit:

- any state is represented as

$$|v\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \qquad \in \mathbb{C}^2$$

- basis set:

$$|v_1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |v_2\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

  indeed,

  - they linearly independent

$$\underbrace{0}_{\text{zero state}} = a_1|v_1\rangle + a_2|v_2\rangle \iff \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{\text{zero state}} = a_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \iff \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$\implies \qquad a_1 = 0 \text{ and } a_2 = 0$$

  - they form a spanning set

$$|v\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = a_1|v_1\rangle + a_2|v_2\rangle$$

- there are infinitely many basis sets, here is another one:

$$|v_1'\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad |v_2'\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

  and (verify!)

$$|v\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{a_1 + a_2}{\sqrt{2}} |v_1'\rangle + \frac{a_1 - a_2}{\sqrt{2}} |v_2'\rangle$$

Aside: why care about different bases?

- when talking about QM/QC we will often identify basis states as outcomes of a measurement of a quantum state
- different measurements will correspond to different sets of potential outcomes
- the coefficients of an arbitrary state will be related to the probabilities of different measurement outcomes

If $|v_1\rangle$ and $|v_2\rangle$ are a basis in a ket space, $\langle v_1|$ and $\langle v_2|$ are then a basis in the corresponding bra space.

$\Longrightarrow$ <u>Inner product</u> defined on a linear vector space takes 2 vectors (**order is important!**) from the space and produces a complex number:

$$\underbrace{(|v\rangle, |w\rangle)}_{\text{inner product of } |v\rangle \text{ and } |w\rangle} \equiv \langle v|w\rangle$$

bra ket

from bracket explains origin of "bra" and "ket" terminology

subject to the following properties:

- linear in the second argument

$$\left(|v\rangle, \sum_i \lambda_i |w\rangle_i\right) = \sum_i \lambda_i \ (|v\rangle, |w\rangle_i)$$

- $$\langle v|w\rangle = \langle w|v\rangle^*$$

- $$\langle v|v\rangle \geq 0\,, \qquad \text{and} \qquad \langle v|v\rangle = 0 \ \Leftrightarrow \ |v\rangle = 0$$

Inner product in $\mathbb{C}^n$ is consistent with a matrix multiplication:

$$|v\rangle = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}, \qquad |w\rangle = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$$

$$\langle v|w\rangle = \langle v|\cdot|w\rangle = \underbrace{(|v\rangle^T)^*}_{1\times n \text{ matrix}} \cdot \underbrace{|w\rangle}_{n\times 1 \text{ matrix}} = \begin{bmatrix} v_1^* & \cdots v_n^* \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \underbrace{v_1^* w_1 + \cdots + v_n^* w_n}_{1\times 1 \text{ matrix}} = \sum_i v_i^* w_i$$

$$\boxed{\begin{array}{c} \text{Linear vector space + inner product = inner product space} \\ \Updownarrow \\ \text{Hilbert space of quantum states of a system} \end{array}}$$

$\Longrightarrow$ Important defs:

- $|v\rangle$ and $|w\rangle$ are *orthogonal* if

$$\langle v|w\rangle = 0$$

- $|v\rangle$ is a unit vector if its norm is 1,

$$||\ |v\rangle\ || \equiv \sqrt{\langle v|v\rangle} = 1 = \sqrt{\sum_i |v_i|^2}$$

Note: we can construct infinitely more additional orthonormal states simply *multiplying a basis vector by a phase*:

$$\{\ |v\rangle_1\,,\ |v\rangle_2\,,\ |v\rangle_3\ \} \qquad \Longrightarrow \qquad \{\ e^{i\phi_1}|v\rangle_1\,,\ e^{i\phi_2}|v\rangle_2\,,\ e^{i\phi_3}|v\rangle_3\ \}$$

where ϕ_i are some arbitrary angles, *e.g.*, note

$$\left(e^{i\phi_1}|v\rangle_1, e^{i\phi_3}|v\rangle_3\right) = e^{-i\phi_1}e^{i\phi_3}\left(|v\rangle_1, |v\rangle_3\right) = e^{i(\phi_3-\phi_1)}\cdot\langle v|_1 v_3\rangle = e^{i(\phi_3-\phi_1)}\cdot 0 = 0 \leftarrow \text{still orthogonal}$$

$$\left(e^{i\phi_2}|v\rangle_2, e^{i\phi_2}|v\rangle_2\right) = e^{-i\phi_2}e^{i\phi_2}\left(|v\rangle_2, |v\rangle_2\right) = e^{i(\phi_2-\phi_2)}\cdot 1 = 1 \qquad \leftarrow \text{still unit vectors}$$

Q: When does the phase matter?

A: We will see that there is no way to measure an absolute phase (ϕ in an overall factor of $e^{i\phi}$) of a state. We can, however, measure a phase difference between two states.

eg. Consider two states $|\psi_1\rangle$, $|\psi_2\rangle = e^{i\phi}|\psi_1\rangle$

In any direct independent measurement of $|\psi_1\rangle$ or $|\psi_2\rangle$ seperately, we would find no difference. However, we could design an experiment involving both $|\psi_1\rangle$ and $|\psi_2\rangle$ which could detect that $|\psi_2\rangle$ is phase shifted from $|\psi_1\rangle$.

$\Rightarrow$ In any calculation, we can set the absolute phase once, (but not multiple times).

A single qubit

- 2 basis states of a qubit (with respect to a measurement along z-axis — also known as *computational basis*) are

$$\underbrace{|0\rangle}_{\text{a spin-up state}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \underbrace{|1\rangle}_{\text{a spin-down state}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

these roughly correspond to *classical* bit states of 0 and 1

- unlike classical bits, a quantum bit (a qubit) can be in a *linear superposition* of the basis states,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \equiv \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

where α and β are arbitrary complex numbers, subject to constrain that $|\psi\rangle$ is a unit vector:

$$1 = ||\ |\psi\rangle\ ||^2 = |\alpha|^2 + |\beta|^2 = 1$$

- A measurement of a qubit irreversibly transforms it into a classical bit with value 0 or 1. Sometimes this is referred to as collapse of the quantum state due to the measurement. i.e. If we measure the state

- for a general state $|\psi\rangle$ measuring the orientation of the spin along z-axis will produce a result with probabilities

to $|0\rangle$

$$p^z_{\text{spin-up}} = |\alpha|^2 = \alpha^*\alpha\,, \qquad p^z_{\text{spin-down}} = |\beta|^2 = \beta^*\beta$$

tic,

- since a measurement result is always either up or down, it must be

$$1 = p^z_{\text{spin-up}} + p^z_{\text{spin-down}} \qquad \Longleftrightarrow \qquad |\alpha|^2 + |\beta|^2 = 1$$

- for a general state $|\psi\rangle$ measuring the orientation of the spin along z-axis will produce a result with probabilities

$$p^z_{\text{spin-up}} = |\alpha|^2 = \alpha^*\alpha\,, \qquad p^z_{\text{spin-down}} = |\beta|^2 = \beta^*\beta$$

- since a measurement result is always either up or down, it must be

$$1 = p^z_{\text{spin-up}} + p^z_{\text{spin-down}} \qquad \Longleftrightarrow \qquad |\alpha|^2 + |\beta|^2 = 1$$

Physically a classical bit can be realized as a charge pulse (in a CPU) or a magnetic moment (in a hard drive). Similarly, a qubit could be realized as any 2-state quantum system. eg. Right or Left polarized photon (part of quantum optics) or the projection of an electron spin along the z-axis (which can only be $+\hbar/2$ or $-\hbar/2$). Realizations of qubits as electron spin are the field of spintronics. To connect with an actual realization of a qubit, we will often use an electron spin along z where

$|0\rangle \Leftrightarrow$ spin-up $\Leftrightarrow$ $+\hbar/2$ projection along z-axis

$|1\rangle \Leftrightarrow$ spin-down $\Leftrightarrow$ $-\hbar/2$ projection along z-axis

- what about the basis states

$$|v_1'\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}, \qquad |v_2'\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix} \qquad ?$$

$\Longrightarrow$ these are identified as $|+\rangle$ and $|-\rangle$ qubit states, associated with measurement of its spin orientation along x-axis

$$|v_1'\rangle \equiv |+\rangle = \frac{1}{\sqrt{2}}\Big(|0\rangle + |1\rangle\Big), \qquad |v_2'\rangle \equiv |-\rangle = \frac{1}{\sqrt{2}}\Big(|0\rangle - |1\rangle\Big)$$

so that when we have

$$|\psi\rangle = \alpha_+|+\rangle + \beta_-|-\rangle, \qquad \alpha_+ = \frac{\alpha+\beta}{\sqrt{2}}, \quad \beta_- = \frac{\alpha-\beta}{\sqrt{2}}$$

$$= \alpha_+\left(\tfrac{1}{\sqrt{2}}|0\rangle + \tfrac{1}{\sqrt{2}}|1\rangle\right) + \beta_-\left(\tfrac{1}{\sqrt{2}}|0\rangle - \tfrac{1}{\sqrt{2}}|1\rangle\right) = \overbrace{\frac{\alpha_+ + \beta_-}{\sqrt{2}}}^{\alpha}|0\rangle + \underbrace{\frac{\alpha_+ - \beta_-}{\sqrt{2}}}_{\beta}|1\rangle$$

$$p_{\text{spin}-(+\text{x})} = |\alpha_+|^2, \qquad p_{\text{spin}-(-\text{x})} = |\beta_-|^2$$

Note:

$$p_{\text{spin}-(+\text{x})} + p_{\text{spin}-(-\text{x})} = \left|\frac{\alpha+\beta}{\sqrt{2}}\right|^2 + \left|\frac{\alpha-\beta}{\sqrt{2}}\right|^2 = \frac{\alpha\alpha^* + \alpha\beta^* + \beta\alpha^* + \beta\beta^*}{2}$$

$$+\frac{\alpha\alpha^* - \alpha\beta^* - \beta\alpha^* + \beta\beta^*}{2} = \alpha\alpha^* + \beta\beta^* = |\alpha|^2 + |\beta|^2 = p^z_{\text{spin}-\text{up}} + p^z_{\text{spin}-\text{down}} = 1$$

- two qubit states $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ (that differ by an overall phase γ) are observationally equivalent;

  NOTE:

$$\text{the same}: \qquad |\psi\rangle_1 \equiv \frac{1}{\sqrt{5}}(|0\rangle + 2|1\rangle) \qquad \text{and} \qquad |\psi\rangle_2 \equiv \frac{e^{i\gamma}}{\sqrt{5}}(|0\rangle + 2|1\rangle)$$

$$\text{not the same}: \qquad \frac{1}{\sqrt{5}}(|0\rangle + 2|1\rangle) \qquad \text{and} \qquad |\psi\rangle_3 \equiv \frac{1}{\sqrt{5}}(|0\rangle + e^{i\gamma}2|1\rangle) \qquad \text{for any } \gamma \neq 0$$

  Interestingly:

  while measuring spin along z-axis

$$p^z_{\text{spin-up}}(|\psi\rangle_1) = p^z_{\text{spin-up}}(|\psi\rangle_2) = p^z_{\text{spin-up}}(|\psi\rangle_3) = \frac{1}{5}$$

  measuring spin along x-axis

$$p_{\text{spin-(+x)}}(|\psi\rangle_1) = \left|\frac{\frac{1}{\sqrt{5}} + \frac{2}{\sqrt{5}}}{\sqrt{2}}\right|^2 = \frac{9}{10}\,, \qquad p_{\text{spin-(+x)}}(|\psi\rangle_2) = \left|\frac{\frac{e^{i\gamma}}{\sqrt{5}} + \frac{2e^{i\gamma}}{\sqrt{5}}}{\sqrt{2}}\right|^2 = \frac{9}{10}\cdot|e^{i\gamma}|^2 = \frac{9}{10}$$

$$p_{\text{spin-(+x)}}(|\psi\rangle_3) = \left|\frac{\frac{1}{\sqrt{5}} + \frac{2e^{i\gamma}}{\sqrt{5}}}{\sqrt{2}}\right|^2 = \frac{|1+2e^{i\gamma}|^2}{10} = \frac{(1+2e^{i\gamma})(1+2e^{-i\gamma})}{10}$$

$$= \frac{5 + 2(e^{i\gamma} + e^{-i\gamma})}{10} = \frac{5 + 4\cos\gamma}{10} \neq p_{\text{spin-(+x)}}(|\psi\rangle_1)$$

$\Longrightarrow$ $|\psi\rangle_1$ and $|\psi\rangle_3$ are observationally distinct

- possible orthonormal basis sets of a single qubit Hilbert space:

$$\{\ |0\rangle\,,\ |1\rangle\ \} \qquad \text{and} \qquad \{\ |+\rangle\,,\ |-\rangle\ \}$$

Bloch sphere representation of a single qubit

An arbitrary qubit state is

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1$$

$\Longrightarrow$

$$|\alpha| = |\cos\frac{\theta}{2}|\,, \qquad |\beta| = |\sin\frac{\theta}{2}|\,, \qquad \theta \in [0, \pi]$$

$\Longrightarrow$

$$\alpha = e^{i\gamma}\cos\frac{\theta}{2}\,, \qquad \beta = e^{i\gamma}e^{i\phi}\sin\frac{\theta}{2}\,, \qquad \phi \in [0, 2\pi)$$

where ϕ is a *relative* phase between α and β

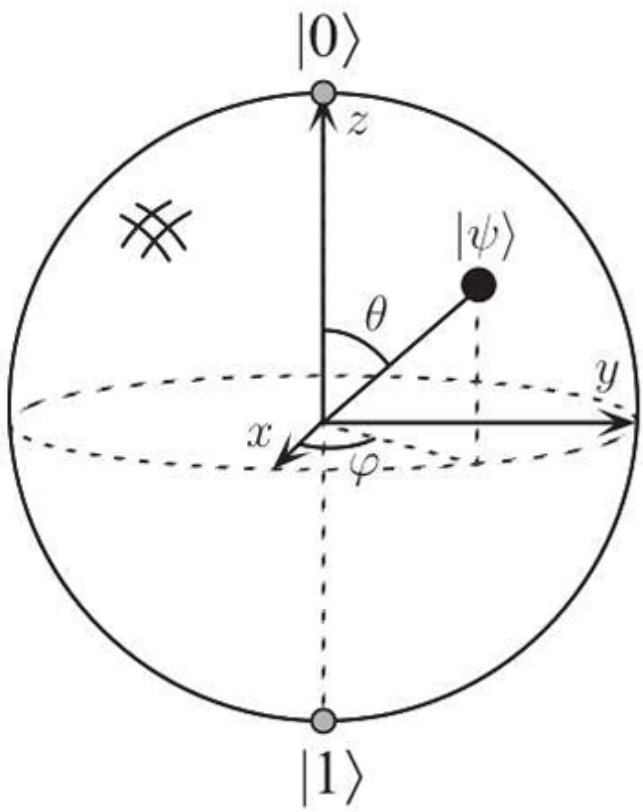
$\Longrightarrow$ overall phase γ is never observable, so we set $\gamma = 0 \Longrightarrow$

$$|\psi\rangle = \cos\frac{\theta}{2}\ |0\rangle + e^{i\phi}\sin\frac{\theta}{2}\ |1\rangle$$

We can visualize $|\psi\rangle$ and a vector

$$\vec{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$$

on a unit sphere:



Note that $|-\rangle$ is oriented along the $-x$ direction

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \implies \theta = \frac{\pi}{2}, \phi = \pi \implies \vec{n} = (-1, 0, 0)$$

Try out a Bloch-sphere demo at:

https://www.st-andrews.ac.uk/physics/quvis/simulations_html5/sims/blochsphere/blochsphere.html